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Optimal Control Strategy for Photovoltaic Shading Devices in Vertical Facades of Buildings

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Abstract

Building energy consumption accounts for a significant portion of total society energy use, and photovoltaic technology is being rapidly deployed across the construction sector. In order to improve the efficiency with which photovoltaic shading devices capture solar energy, a numerical calculation model for the ideal tilt angle of these devices is constructed in this study. This model is based on clear-sky solar radiation calculation algorithms and solar radiation resources across different latitudes. In order to maximize solar radiation collection, an ideal control strategy for photovoltaic shading devices on buildings with varied orientations at different latitudes and in different months is derived through numerical simulations. The findings demonstrate that the building's orientation has a significant role in determining how well photovoltaic shading systems use solar energy. In winter, the ideal tilt angle for south-facing facades increases by 10° for every 10° increase in latitude. And for every 25° rise in latitude, the ideal tilt angle increases by only around 10° in summer. By applying optimal regulatory strategies, solar radiation consumption efficiency of roughly 65% can be attained, providing a reference basis for boosting power generating efficiency and building energy saving.

Keywords: photovoltaic shading devices; clear-sky solar radiation; optimal control strategy; energy utilization

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1. Introduction

Global warming represents one of the most urgent environmental challenges confronting humanity in the present day. The massive carbon dioxide (CO_2) emissions generated by the combustion of fossil fuels are the main human-driven cause of this climate phenomenon [1]. In recent years, there has been a substantial year-on-year increase in electricity generation from renewable sources such as solar and wind power, with their capacity accounting for nearly 85% of the net electricity growth demand in 2022 [2]. Globally, the energy used by buildings accounts for a significant proportion of total energy consumption. In order to achieve the goal of net-zero carbon emissions for all buildings by 2050 [3], the international community is encouraging the use of renewable energy in order to optimize energy systems and promote sustainable development [4,5]. Consequently, the implementation of renewable energy in the construction sector has emerged as an unavoidable trend in the global development of the building and energy industries.

Solar energy is widely recognized as a free, clean and unlimited renewable energy source that is environmentally friendly [6]. Among the wide range of building-integrated photovoltaic applications, photovoltaic shading devices (hereafter referred to as PVSDs) provide notable benefits. PVSDs integrate conventional shading functionalities with photovoltaic energy generation technology, offering an innovative and sustainable green energy solution for building facades. They effectively regulate indoor temperatures and reduce cooling loads, while efficiently converting solar energy into electricity [7,8]. As societal awareness of the need to replace conventional energy sources with renewable energy grows, more and more research is focusing on optimizing the utilization of solar radiation resources through shading photovoltaic panels.

Taveres-Cachat et al. found that PVSDs with fewer louvers offer greater flexibility in adjusting tilt angles, which can significantly enhance performance. This confirms the pivotal role of tilt angles in PVSD optimization [9]. Ito and Lee used multi-objective optimization methods to optimize the shape of PVSDs featuring flexible solar panels and curved louvre arrays, thereby reducing overall building energy consumption further [10]. Baghoolizadeh conducted energy simulations across five European cities with diverse climates, demonstrating that correctly angled, sized and positioned shading curtains can effectively conserve energy. Compared to static PVSD configurations, appropriately optimized tilt angles yield electricity cost savings ranging from 17% to 30% [11]. Zhang's numerical simulations revealed that installing solar photovoltaic shading panels at a 30-degree tilt angle on south-facing façades in Hong Kong optimizes power generation efficiency and maximizes energy-saving potential [12]. Jing et al. employed an inverse distance weighting model, the Collares-Pereira and Rabl time-of-day radiation model, and an oblique radiation model to determine China's annual total global radiation at 3097–7311 megajoules per square metre, diffuse radiation at 495–3036 megajoules per square meter, and an optimal tilt angle range of 14.5° to 49.1°. Furthermore, the optimum tilt angle exhibits a positive correlation with latitude [13]. Li developed a simulation model based on EnergyPlus 8.5 to analyze net electricity consumption for PVSDs with varying tilt angles and widths across five thermally representative Chinese cities. This identified optimal annual and monthly tilt angles. The results indicate that south-facing PVSDs offer the greatest energy-saving potential [14]. Akbari Paydar et al. compared dynamic and static PVSD applications on a south-facing apartment building in Tehran using EnergyPlus 8.5 and MATLAB 7.10.0. They found that adjusting the angle and width of PVSDs twice annually effectively reduced building energy consumption [15]. Sun et al. simulated Hong Kong's meteorological conditions and determined that south- and southwest-facing shading-optimized PVSDs achieve higher annual power generation, with south-facing modules reaching maximum output at a 10° tilt angle [16]. Svetozarevic et al. proposed a dynamic photovoltaic fence structure for buildings, utilizing hybrid material actuators. Compared to static PV fence designs, this approach increased electricity yield by 50%, meeting 115% of an office building's energy demand [17]. Han et al. conducted a multi-objective optimization study using PVSDs parameters including tilt angle, orientation, width of the bPV sunshades, wall reflectivity, window reflectivity, and window-to-wall ratio. The research concluded that the optimal orientation for Hong Kong's sunshades is due south with a tilt angle of 35°. From an energy-saving perspective, a smaller tilt angle and south-facing sunshade width often yield greater energy-saving benefits [18]. Wang et al. studied multi-story dynamic PVSDs. PVSDs installed on the south façade achieved an annual power generation of 1524.63 kWh at the optimal hourly rotation angle, with an average efficiency of 0.11 kWh/m^2/day, while effectively avoiding shading between floor installations [19]. Wang et al. innovatively proposed a dynamic PV-integrated lighting frame system. Compared to fixed PV shading systems, this system increased annual electricity generation by 12% and reduced net energy consumption by 3.56% [20]. Ye and Zhu et al. proposed a multi-objective optimization framework to maximize photovoltaic potential while minimizing PV area. Utilizing GIS-based spatio-temporal analysis and optimization methods, they compared three scenarios: parallel to the horizontal ground; a fixed tilt angle; and real-time rotation to maintain the vertical alignment of the PV surface with solar radiation. They concluded that, under real-time rotation conditions, PVSDs with a width of 0.7 m achieve an annual power generation of 0.861 GWh [21].

The above researchers have conducted extensive investigations into optimizing the energy utilization efficiency of photovoltaic shading systems. Nonetheless, it is important to acknowledge that more than 90% of current studies concentrate exclusively on south-facing orientations. Given the diversity of building construction styles and orientations, this study represents the initial exploration into azimuth dimensions, examining optimal inclination angle configurations for non-south-facing PVSDs positioned at 30/60 degrees west/east of south. It addresses a research void in the optimization of PVSDs for non-south-facing orientations. Secondly, the computation of solar radiation intensity depends on meteorological parameters and employs specialized software such as EnergyPlus and MATLAB for forecasting and simulation. This typically necessitates that designers possess a high level of professional expertise. This study utilizes an Excel-based computational model derived from the Hottel model, facilitating dip angle calculations without the need for specialized software and reducing the barriers to engineering implementation. Thirdly, research primarily focuses on particular cities or climate zones without developing models to account for tilt angle variations across different latitudes. Consequently, conclusions lack general applicability and demonstrate deficiencies in practical implementation. This is especially true in mainland China, where integrated shading and photovoltaic systems are comparatively recent developments. Current research thus provides inadequate guidance.

This study examined a highly adaptable paradigm for harnessing solar radiation. Aiming to optimize the capture of direct solar radiation, the model considers multiple multidimensional factors, including building orientation, geographical latitude, and monthly fluctuations. Through comparison study, a monthly regulation strategy is proposed that ensures the efficiency of solar radiation utilization exceeds 98% of the optimal tilt angle. This can be achieved through manual or straightforward mechanical adjustments once a month or quarterly, significantly lowering the operational threshold for PVSDs. This research promotes renewable energy production, decreases building energy usage, expedites the attainment of net-zero energy building objectives, and advances sustainable building development.

2. Materials and Methods

2.1. Model Description

Existing studies utilize standard meteorological data; however, actual weather conditions deviate from these parameters on a daily basis. This discrepancy may lead to optimal tilt angles that do not effectively maximize solar radiation utilization on clear days (e.g., when a genuinely clear day is recorded as overcast in meteorological data). This paper examines a comprehensive clear-sky model, presuming the absence of overcast days throughout the year. This methodology is justified due to the fact that overcast conditions result in reduced total solar radiation intensity, with almost all of the radiation being diffuse. Since diffuse radiation is isotropic, the quantity of radiation captured at any given angle remains uniform, thereby negating the necessity for an optimal tilt angle. Therefore, this paper presumes a completely clear sky to analyze the optimal inclination angle for each month. This method optimizes the capture of direct solar radiation and accounts for deviations resulting from discrepancies between actual weather conditions and meteorological parameter data. The clear-sky factor is established at 0.7 rather than 1, reflecting the most probable occurrence of clear weather conditions, in which both direct and diffuse radiation are present. As this is a clear-sky model, the intensity of dispersed radiation remains relatively low, averaging 12.6% of total annual radiation, with a maximum not exceeding 17.5%. Therefore, the assumption of isotropy is employed to facilitate calculations.

In this study, the optimal tilt angle is investigated by calculating the solar radiation incident on the surface of PVSDs. To optimize the accumulation of solar radiation, it is assumed that there is no self-shading between floors, no shading from other buildings, and the effect of PV module temperature is neglected.

The calculation time increment is designated as one hour, presuming that solar radiation intensity remains unchanged throughout this interval. The radiation intensity at the one-hour mark is utilized as the instantaneous heat gain, expressed in watts (W). This quantity is subsequently multiplied by the duration of one hour to determine the total radiation accumulated during that period, expressed in kilowatt-hours (kWh). The elevation angle is utilized in establishing the angular range for sunrise and sunset. If the elevation angle is below 0°, it signifies the interval preceding sunrise or following sunset; if the elevation angle exceeds 0°, it denotes daytime.

Five representative regions across low to high latitudes in China were selected: Haikou (20° N), Kunming (25° N), Nanjing (32° N), Beijing (40° N) and Harbin (45° N). These regions were selected due to their notable variations in atmospheric transparency and other related parameters. Owing to the isotropic characteristics of sky-scattered radiation, the quantity received by photovoltaic panels per unit area remains predominantly unchanged regardless of the panels' inclination angle. Consequently, diffuse radiation exerts a negligible influence on PVSD energy collection. Clear-sky direct solar radiation thus functions as the principal energy source for PVSDs, with its magnitude affected by factors such as geographic latitude, solar altitude angle, and solar azimuth angle. In this study, the Hottel Model [22] was utilized to calculate clear-sky solar radiation, incorporating specific data such as building orientation to optimize results. A model was developed to calculate hourly solar radiation intensity throughout the year and serves as the analytical basis. Figure 1 shows the tilt angle and orientation of PVSDs.

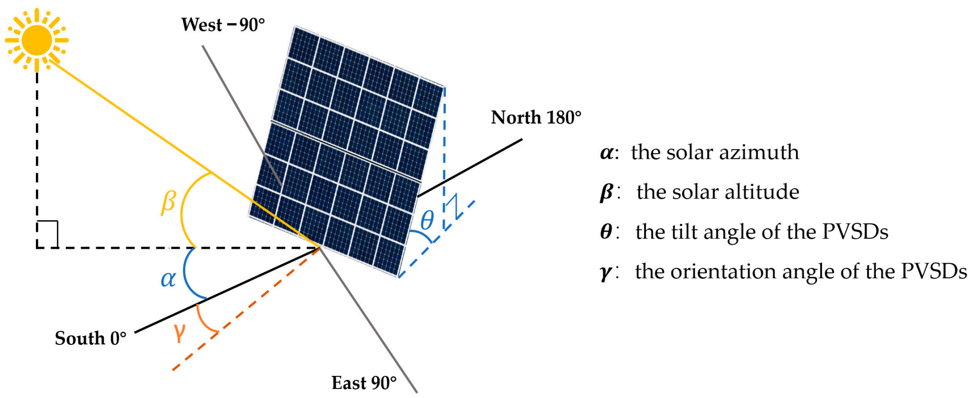


Figure 1. Solar Panel Tilt Angle and Orientation. (Solid lines indicate direction; dashed lines represent angle reference lines.).

2.2. Clear-Sky Radiation Optimization Calculation Model

The intensity of solar radiation outside the atmosphere is calculated according to Equation (1), enabling the determination of the solar radiation flux G_{on} beyond the atmosphere at any latitude and at any given time.

$$G_{on} = G_{sc}\left[1 + 0.033\cos\left(\frac{360n}{365}\right)\right], \quad (1)$$

where G_{sc} is solar constant and its value is 1374 W/m^2, n is the number of the calculating day counting from 1 January. For example, n is 15 for 15 January, and n is 32 for 1 February.

Clear-sky solar radiation transmittance τ_b is calculated using Equation (2).

$$\tau_b = a_0 + a_1 exp(-\frac{k}{cos\theta_z}), \tag{2}$$

The coefficient can be determined by the following equations:

$$a_0 = r_0 \times \left[0.4237 - 0.00821(6 - A)^2\right]$$

$$a_1 = r_1 \times \left[0.5055 + 0.00595(6.5 - A)^2\right]$$

$$k = r_k \times \left[0.2711 + 0.01858\,(2.5 - A)^2\right]$$

The correction factors (r_0, r_1, r_k) are determined by the climate type, such as tropical climate, mid-latitude summer climate, cold belt summer climate and mid latitude winter climate. The specific values range from 0.95 to 1.03, which can be found in the Hottel Model [22].

The declination angle δ is calculated using Equation (3).

$$\delta = 23.45 \times \sin\left(360 \times \frac{284 + \mathrm{n}}{365}\right), \tag{3}$$

The true solar time is calculated using Equation (4). The second term is positive if the location is in the Eastern Hemisphere; conversely, it is negative if the location is in the Western Hemisphere.

$$T = T_m \pm \frac{L - L_m}{15} + \frac{e}{60'} \tag{4}$$

where T is the local true solar time, h; T_m is the average solar time (or the standard time) in the corresponding time zone, h; L is the local meridian longitude, deg; L_m is the longitude of the central meridian of the corresponding time zone, deg; e is the time difference, min.

The time angle h is calculated using Equation (5).

$$h = 15 \times (T - 12) \tag{5}$$

The solar altitude β is calculated using Equation (6).

$$\beta = \sin^{-1}(\cos\varphi \cos h \cos\delta + \sin\delta \sin\varphi), \tag{6}$$

where φ is the geographic latitude, deg; h is the time angle, deg.

The solar azimuth α is calculated using Equation (7).

$$\alpha = \sin^{-1}(\cos\delta \cos h) \div \cos\beta, \tag{7}$$

The direct solar radiation flux I_{dir} is calculated using Equation (8).

$$I_{dir} = G_{on} \times \tau_b, \tag{8}$$

The diffuse solar radiation flux I_{dif} is calculated using Equation (9).

$$I_{dif} = (0.271 - 0.294\tau_b) \times G_{on} \times \sin\beta, \tag{9}$$

The total amount of solar radiation received by the PVSDs is calculated using Equation (10).

$$Q = I_{dir} \times \cos(90^\circ - \beta - \theta) \times \cos|\alpha - \gamma| + I_{dif}, \tag{10}$$

where θ is the tilt angle of the PVSDs (as shown in Figure 1), °; γ is the orientation angle of the PVSDs, °.

2.3. Model Validation

In this model, the solar altitude "β" and the solar azimuth "α" are obtained from the classical Hottel Model [22], which is widely acknowledged.

The accuracy of the clear-sky model was verified through the use of direct and diffuse solar radiation parameters for the Beijing region, obtained from the SWERA (Solar and Wind Energy Resource Assessment) meteorological parameter dataset. This weather parameter file includes data sourced from the United Nations Environment Programme's satellite-derived datasets, which are available for download from the EnergyPlus software's weather data portal. Data pertaining to clear-sky conditions was extracted from the SWERA meteorological parameter file. The model's computed results were subsequently compared with the solar radiation data contained in the file, and the findings are displayed in Table 1. The discrepancy between the solar radiation intensity calculated by the model and the data recorded in the meteorological file was less than 10%. Therefore, the established clear-sky model demonstrates a high degree of accuracy.

Table 1. Calculation verification of the solar radiation intensity.

Date	Time	Solar Radiation	Intensity Calculated by Clear-Sky Model (W/m^2)	Intensity in SWERA (W/m^2)	Error (%)
6 March	10:00	direct solar radiation	705	639	10.3%
		total solar radiation	804	768	4.7%
	12:00	direct solar radiation	771	729	5.8%
		total solar radiation	876	895	−2.1%
	14:00	direct solar radiation	724	669	8.2%
		total solar radiation	825	842	−2.0%
23 September	10:00	direct solar radiation	709	671	5.7%
		total solar radiation	807	824	−2.1%
	12:00	direct solar radiation	764	750	1.9%
		total solar radiation	868	933	−7.0%
	14:00	direct solar radiation	725	692	4.8%
		total solar radiation	825	867	−4.8%
31 December	10:00	direct solar radiation	541	534	1.3%
		total solar radiation	618	604	2.3%
	12:00	direct solar radiation	650	705	−7.8%
		total solar radiation	736	804	−8.5%
	14:00	direct solar radiation	573	583	−1.7%
		total solar radiation	653	677	−3.5%

To validate the precision of the established model in estimating the amount of solar radiation captured by PVSDs, two specific conditions were considered: one in which the PVSDs are oriented perpendicular to the direct solar radiation and another in which they are aligned parallel to the direct solar radiation. These two conditions are shown in Figure 2.

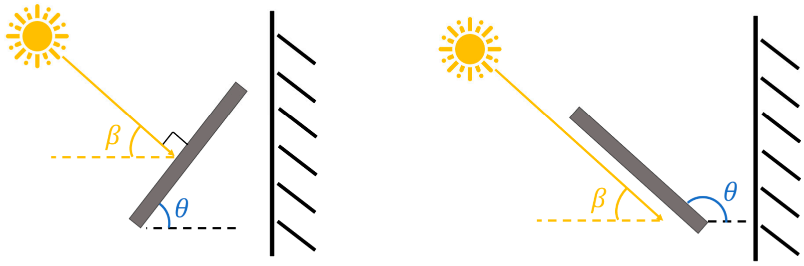


Figure 2. Two specific conditions used to validate the model. (**a**) PVSDs perpendicular to direct solar radiation; (**b**) PVSDs parallel to direct solar radiation.

As shown in Figures 1 and 2, if the PVSDs is perpendicular to the direct solar radiation, then $\alpha = \gamma$, $\theta = 90^\circ - \beta$. At this stage, the PVSDs are capable of receiving the entire quantity of both direct and diffuse solar radiation, allowing the solar radiation efficiency to reach 100%. If the PVSDs is parallel to the direction of direct solar radiation, then $\alpha = \gamma$, $\theta = 90^\circ + \beta$. Currently, the PVSDs exclusively receive diffuse radiation. Choose the calculation parameter values for Beijing at 12:00 on January 1, as presented in Table 2. The theoretical value corresponds to the sum of the direct radiation intensity and the diffuse radiation intensity.

Table 2. Calculation verification under specific conditions.

The Angles	Direct Solar Radiation Intensity (W/m^2)	Diffuse Solar Radiation Intensity (W/m^2)	PVSDs to Direct Solar Radiation	θ (°)	Calculated Value (W/m^2)	Theoretical Value (W/m^2)
$\beta = 26.95^\circ$ $\alpha = 2.42^\circ$ $\gamma = 2.42^\circ$	650	87	perpendicular	63.05	$650 \times cos(90^\circ - 26.95^\circ - 63.05^\circ) \times cos0^\circ + 87 = 737$	650 + 87 = 737
			parallel	116.95	$650 \times cos(90^\circ - 26.95^\circ - 116.95^\circ) \times cos0^\circ + 87 = 87$	87

2.4. Parametric Performance Design of PVSDs

Parametric performance design methods combine parametric modeling with performance evaluation, offering non-specialists a visual arsenal for the optimization and assessment of design proposals [23]. Based on the solar radiation calculation model optimized in Section 2.2, a parametric design framework was developed. The model's input parameters consist of region-specific fixed factors, such as solar altitude angle and solar azimuth angle, as well as dynamic variables, including photovoltaic panel inclination angle and building orientation angle. Through iterative calculations of tilt angles, the monthly optimal tilt angles and the corresponding adjustment ranges for distinct regions were established. This guarantees that the efficiency of solar radiation utilization within the tilt angle range differs by no more than 2% from that attained at the optimal tilt angle. The method ensures that the angles within the optical range achieve a solar radiation utilization efficiency of over 98% at the optimal tilt angle. During the parametric design process for photovoltaic panel tilt angles, consideration can also be given to the relationships among parameters and design schemes, thereby improving both the quality and efficacy of the design. The optimal regulation strategy design phases for PVSDs are depicted in Figure 3.

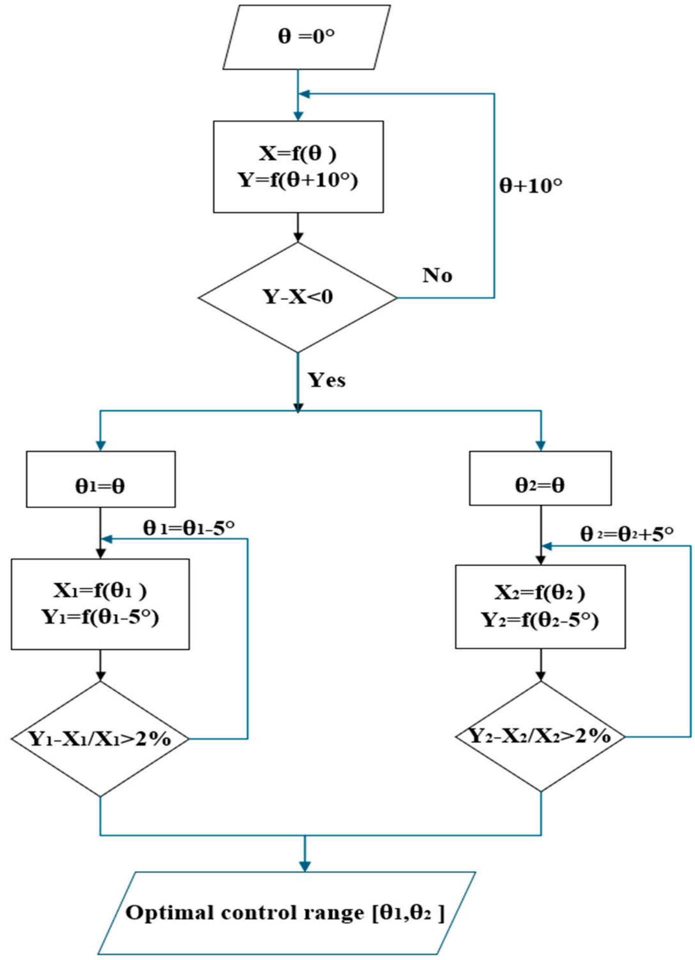


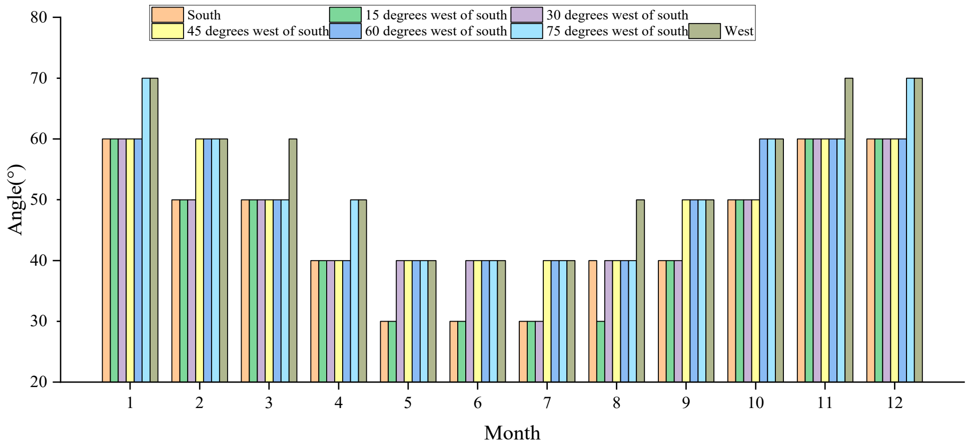
Figure 3. Optimal Regulation Strategy Design Flowchart for PVSDs.

3. Results and Discussions

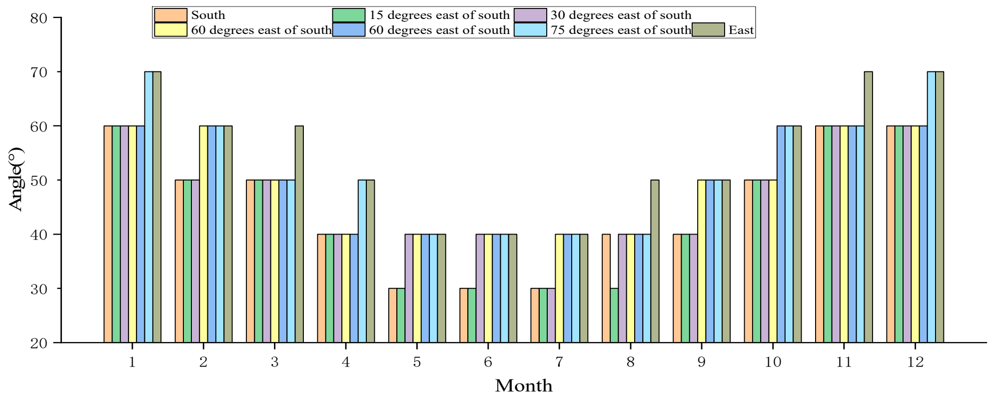
3.1. Variation Patterns of Optimal Tilt Angles for PVSDs Across Different Building Orientations in Nanjing

Nanjing, the provincial capital of Jiangsu, is situated near Shanghai and lies within a climatic zone marked by scorching summers and frigid winters. As an economically advanced city characterized by high building density, it presents substantial potential for the integrated deployment of photovoltaic shading devices on building facades. Therefore, Nanjing was chosen as a representative city to examine the optimal tilt angles of PVSDs across various orientations.

To maximize the collection of direct solar radiation, orientations were selected at 15° intervals from true south as study subjects. Numerical calculations determined the optimal tilt angles for PVSDs across different orientations in Nanjing for each month (as shown in Figure 4). Based on the results, all orientations exhibit a pattern of "higher tilt angles in winter and lower in summer." For buildings facing different orientations, the optimal tilt angle in winter ranges from 60° to 70°, while the optimal tilt angle in summer ranges from 30° to 40°. This trend opposes the seasonal variation in solar altitude angle, validating the core design logic that "tilt angle compensates for solar altitude angle." As the deviation from true south increases (eastward or westward), the optimal tilt angle generally shows an upward trend. This occurs because east- or west-facing buildings primarily receive oblique radiation in the morning or afternoon, allowing photovoltaic panels to be more vertical, thereby minimizing radiation loss. The difference in optimal inclination angle between east and west orientations at the same deviation angle is insignificant. This suggests that, under clear-sky conditions exclusively, the optimal photovoltaic inclination angles for east and west orientations demonstrate symmetry across various regions, offering a consistent reference for the design of PV shading devices on building east and west facades.



(**a**)



(**b**)

Figure 4. Optimal tilt angles for different orientations in Nanjing across various months: (**a**) Southwest; (**b**) Southeast.

3.2. Optimal Tilt Angles for PVSDs Across Various Latitudes

First, with an emphasis on the south-facing orientation, the utmost total solar radiation received by PVSDs each month across five regions at various inclination angles is determined. The results, presented in Figure 5, demonstrate the optimal tilt angles for south-facing panels in each region. The figure demonstrates that tilt angles tend to increase with latitude throughout the year. Specifically, in Haikou, the ideal slope angle for photovoltaic panels is approximately 30° from April to September and around 50° during the winter months. In Kunming, the optimal inclination angle for photovoltaic panels is approximately 30° from April to August, increasing by roughly 10° relative to Haikou during winter, reaching approximately 60°. In Beijing, the optimal tilt angle is approximately 40° from April to August, increasing to approximately 70° during winter. In Harbin, the overall inclination angle for each month aligns with that of Beijing, except for an increase of approximately 10° during March and April. In summary, for installations facing south, the optimal winter inclination angle increases by approximately 10° for every 10° increase in latitude. The ideal summer tilt angle is minimally influenced by latitude, increasing by approximately 10° for every 20° rise in latitude.

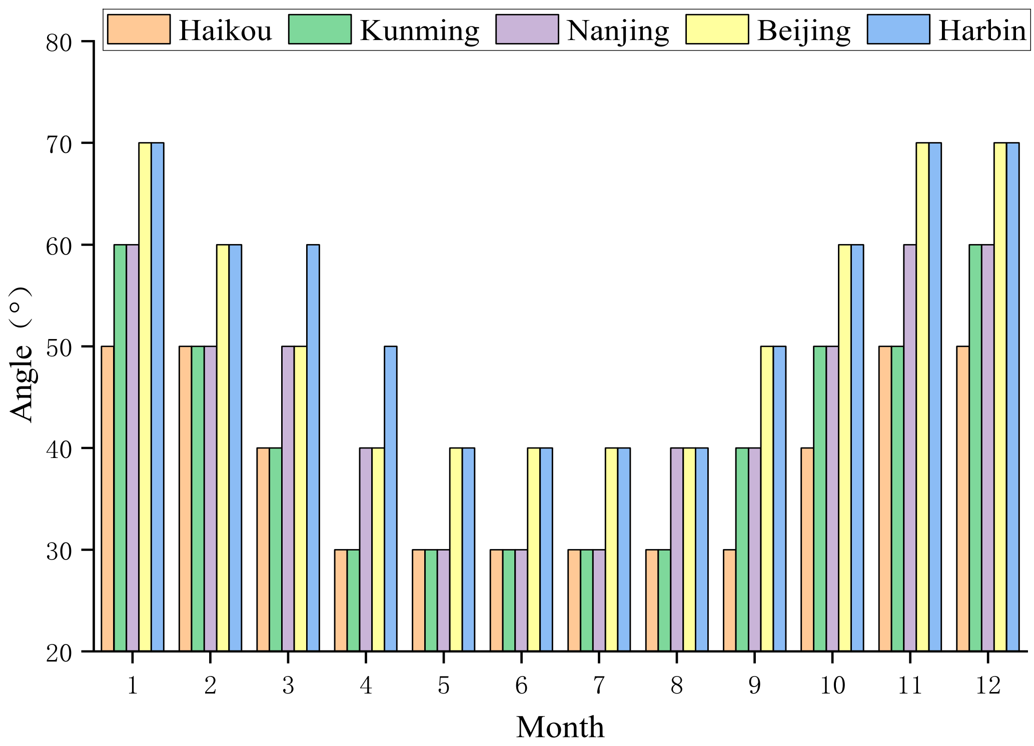
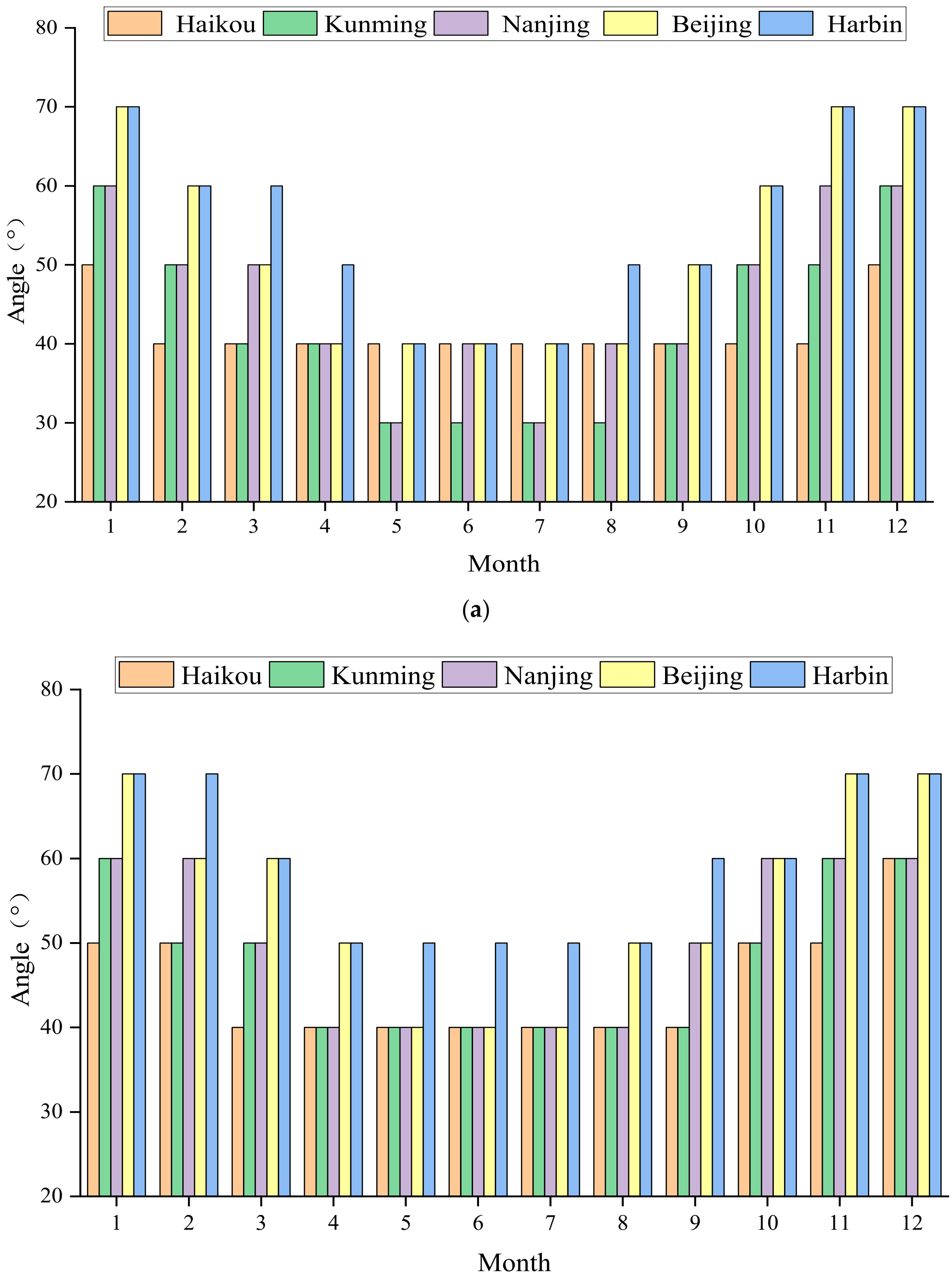


Figure 5. Optimal south-facing tilt angle variation across different latitudes.

Subsequently, the variation in optimal inclination angles across different latitudes for buildings with diverse orientations is examined. Since the optimal inclination angles for east/west orientations display symmetrical patterns, only the southeast orientation is employed as a representative example when examining the effects of latitude. Three building facade orientations were selected: southeast at 30°, southeast at 60°, and due east. The variation patterns of their optimal photovoltaic panel inclination angles are depicted in Figure 6.

As shown in Figure 6, higher latitudes demonstrate larger optical tilt angles throughout all months of the year, with a more significant increase during the winter season. Across all latitudes, the optimal inclination angle for all orientations adheres to the consistent pattern of being higher in winter and lower in summer, which contrasts with the seasonal variation in solar altitude angles. During winter, the solar altitude angle is comparatively low. By increasing the tilt angle of the photovoltaic panels, solar radiation can be directed more vertically towards the panels, thereby reducing reflection losses. In summer, when the solar altitude angle is relatively high, a smaller tilt angle can compensate for this, minimizing the incidence angle of direct solar radiation. For a specified orientation, the optimal inclination angle rises with latitude during the winter season. On average, a 10° increase in latitude is associated with a 10° increase in the optimal tilt angle during winter. This pattern closely mirrors the trend observed for south-facing facades, wherein the optimal winter inclination angle increases by 10° for every 10° increase in latitude. However, for identical orientations, latitude exerts a lesser influence on the optimal inclination angle during summer. For each 25° increase in latitude, the optimal lean angle rises by approximately 10° during the summer.

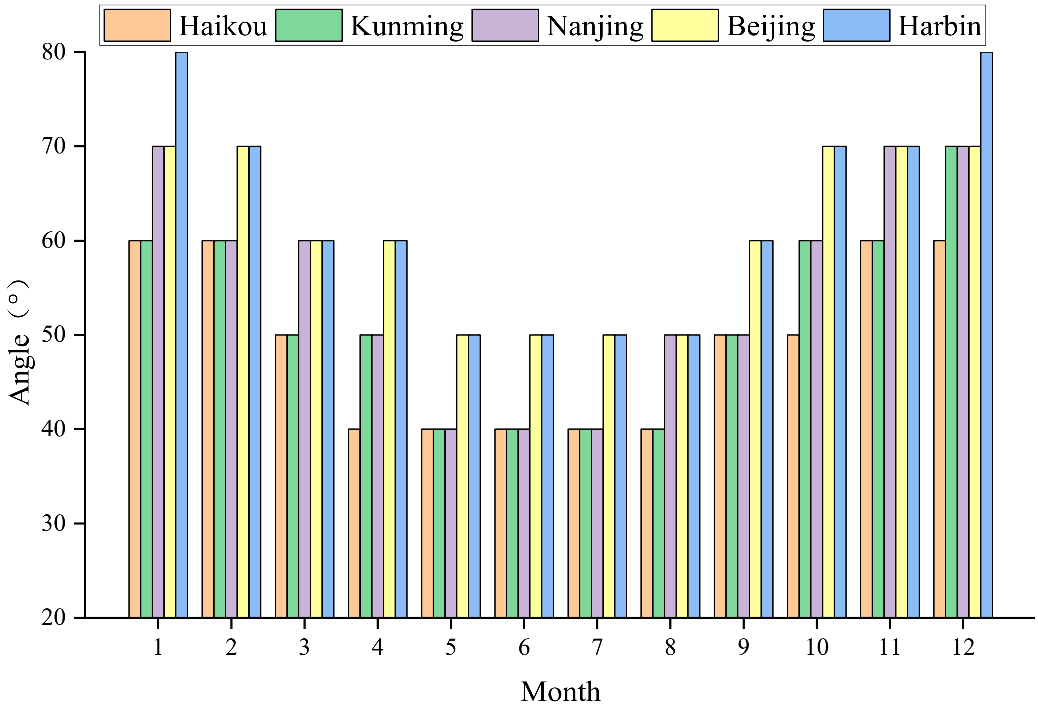
The more a building's orientation deviates from true south, the greater its overall inclination angle throughout the year, with winter fluctuations being more significant. Conversely, the optimal tilt angles for the three principal orientations during summer typically range between 30° and 45°, exhibiting minimal variation among orientations at the same latitude. This phenomenon occurs because the summer sun's altitude angle varies significantly, enabling all orientations to receive substantial direct radiation and necessitating only minor adjustments to fulfill the requirements. During winter, the ideal tilt angles for the three orientations typically vary between 50° and 80°. At the same latitude, the differences in winter inclination angles between orientations are marginally larger than those observed in summer. This is due to the fact that the sun's azimuth during winter is more closely aligned with true south. Consequently, as the orientation changes, a greater inclination angle is necessary to offset radiation losses resulting from variations in azimuth. As outlined in Section 3.1, panels aligned along the east–west axis demonstrate symmetry. Consequently, the optimal inclination angle pattern for southwest-oriented panels is essentially similar to that of southeast-oriented panels.



(**a**)

(**b**)

Figure 6. *Cont*.



(**c**)

Figure 6. Optimal tilt angle for different orientations across latitudes: (**a**) 30 degrees east of south; (**b**) 60 degrees east of south; (**c**) east.

3.3. Efficiency of Solar Radiation Utilization Across Regions with Varying Latitudes and Building Orientations

Based on the previously established clear-sky solar radiation calculation model established earlier and the analysis of optimal tilt angles, a quantitative analysis of solar radiation utilization efficiency was conducted for five representative regions—Haikou (20° N), Kunming (25° N), Nanjing (32° N), Beijing (40° N), and Harbin (45° N)—using the cumulative clear-sky solar radiation amounts as the standard value, representing the total amount of both direct and diffuse solar radiation ($I_{dir} + I_{dif}$). The comprehensive results of the analysis are presented in Table 3.

Table 3. Solar utilization efficiency at optimal tilt angles for cities at different latitudes.

	South	30 Degrees East/West of South	60 Degrees East/West of South	East/West
Haikou	62.63%	57.38%	52.54%	42.85%
Kunming	64.67%	59.38%	53.08%	41.31%
Nanjing	66.90%	61.16%	52.86%	39.93%
Beijing	69.17%	63.25%	53.02%	39.49%
Harbin	70.17%	64.15%	53.27%	39.31%

As shown in Table 3, at their respective optimal tilt angles, orientation markedly influences the efficiency of solar radiation utilization: efficiency steadily declines as the deviation from true south widens, with notable differences in efficiency among various orientations. At the optimal inclination angle, the utilization efficacy of south-facing PVSDs across the five regions varies from 62% to 70%. This orientation enables buildings to receive the most consistent and ample sunlight with minimal periods of radiation loss, making it the optimal choice for PVSD applications. Panels tilted 30 degrees east or west of south generally attain approximately 60% efficiency, with only slight losses during midday hours. Efficiency declines markedly for panels facing 60 degrees east or west of south, with higher-latitude regions (e.g., Beijing, Harbin) generally showing lower efficiency than lower-latitude areas (e.g., Haikou, Kunming). This phenomenon arises because the increased deviation from true south at these orientations significantly diminishes midday solar radiation, thereby causing panels to depend mainly on oblique radiation during morning and afternoon periods. East/West-oriented panels demonstrate the lowest efficacy, ranging from 39% to 43%. This decline becomes more evident with increasing latitude. Considering the limited total radiation collected over the course of the day, further optimization strategies are necessary.

The influence of latitude on the utilization of solar radiation diminishes as the orientation diverges from due south. When the orientation is 60 degrees east or west of south, the solar radiation utilization efficiency in Haikou is 52.54%, representing a difference in less than 1% compared to Harbin's efficiency of 53.27%. This efficiency is significantly lower than the 7.54% observed in the due south orientation. This occurs because the solar azimuth in high-latitude regions tends to be more southerly during winter.

3.4. Efficiency of Solar Radiation Utilization Under Different Regulatory Strategies

The solar radiation utilization rates across various orientations and four tilt angle control strategies are calculated and compared. The four tilt angle control strategies are: (1) horizontally mounted PVSDs; (2) vertically mounted PVSDs; (3) PVSDs fixed at the optimal tilt angle for the entire year; (4) PVSDs fixed at the optimal tilt angle for each month. The results of the calculations are presented in Figure 7.

As shown in Figure 6, irrespective of the building orientation, the monthly optimal tilt control strategy attains the utmost efficiency in solar radiation utilization, markedly surpassing the other three strategies. Under a true south orientation, the efficacy of the monthly optimal tilt ranges from approximately 63% to 70%, while the fixed annual tilt attains roughly 60% to 67%. Horizontal and vertical configurations demonstrate significantly reduced efficiencies.

Under various control strategies, the effectiveness of solar energy utilization varies considerably depending on building orientation. In the south-facing orientation, the efficiency disparity between the monthly optimal tilt angle strategy and the horizontal installation approach can reach up to 34%. At 30° south by east, the disparity in efficacy between the two diminishes to approximately 29%. In the east-facing direction, the utmost efficiency discrepancy further diminishes to 15%. Therefore, the south direction receives the most abundant and uniform solar radiation, and the monthly optimal tilt angle adjustment strategy can precisely correspond to the variations in the solar elevation angle across different months. Under the horizontal/vertical installation strategy, the fixed angle of the flipper results in substantial radiation loss during midday; the east-facing facade only receives morning radiation, and even with optimized tilt, the increase in radiation collection throughout the day remains limited, thereby reducing the efficiency differential.

The efficiency disparity between control strategies and alternative orientations becomes increasingly significant as the building orientation approaches true south: at true south orientation, the efficiency difference between the monthly optimal tilt and horizontal installation ranges from 12% to 24%; at 30° east of south, the gap narrows to 9% to 21%; and at true east orientation, the difference further diminishes to 10% to 13%. South-facing facades receive the most extensive and consistent solar radiation. The optimal tilt angle enhances the capture of peak midday radiation, whereas fixed horizontal or vertical installations experience considerable radiation loss during midday. A true east orientation receives solely morning radiation with a limited overall quantity, thereby reducing the optimization benefits of various strategies and consequently narrowing the efficiency disparity among them.

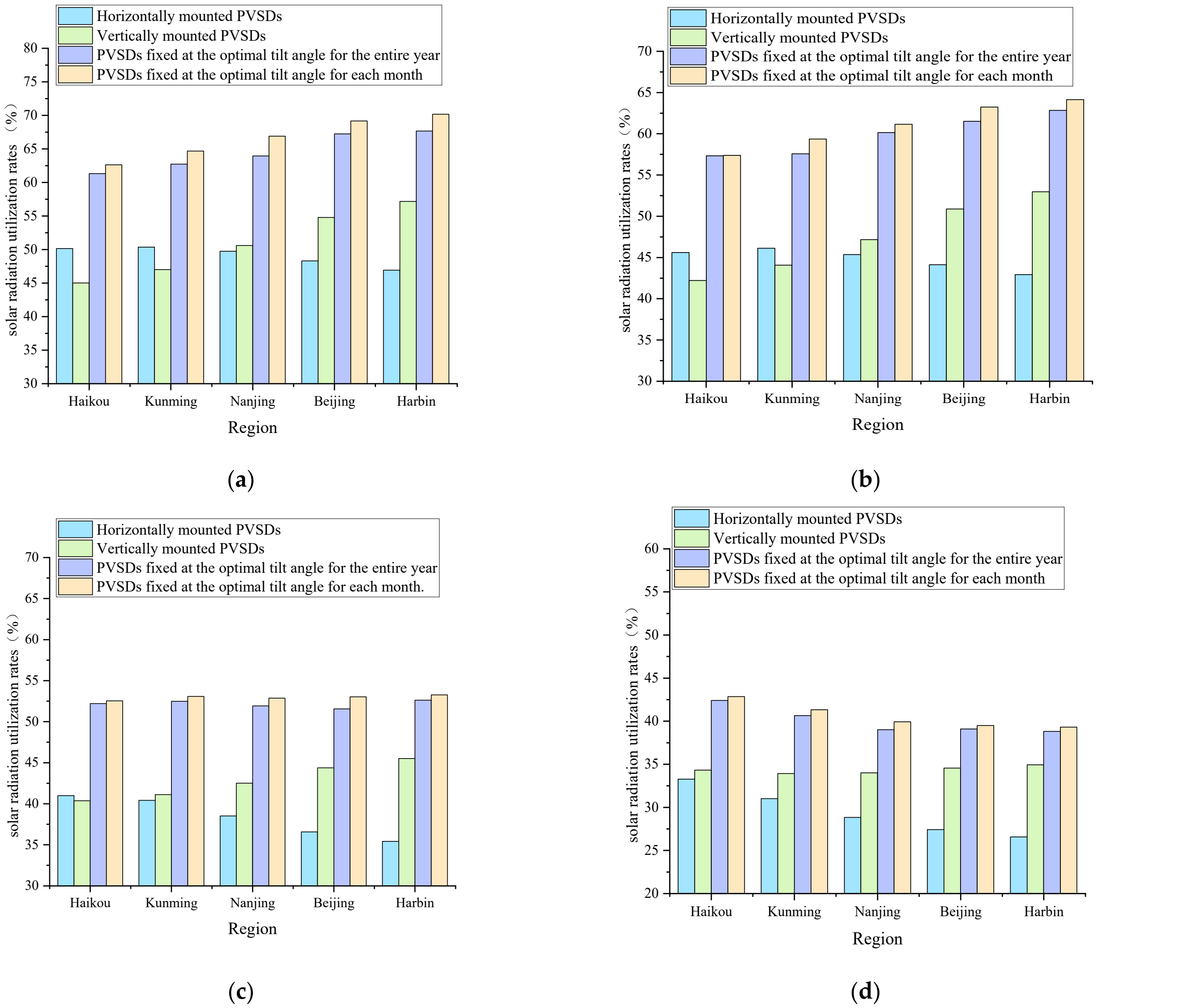


Figure 7. Solar radiation utilization efficiency under different tilt angle strategies of PVSDs: (**a**) south; (**b**) 30 degrees east of south; (**c**) 60 degrees east of south; (**d**) east.

3.5. Optimal Control Strategies for PVSDs

Based on the optimal tilt angles for each month, a parametric performance design methodology is utilized to integrate range data into the calculations. It summarizes the optimal control strategies for PVSDs, ensuring that solar radiation utilization approaches the efficiency attained at the optimal inclination angle, with an error margin below 2%. The primary objective is to facilitate indoor personnel in regularly modifying the optimal inclination angle of photovoltaic panels by offering an optimal control range. Table 4 displays the optimal control strategies for PVSDs across various orientations in Beijing. across various latitudes and orientations. The Appendix A presents the optimal control strategies for cities situated at different latitudes.

Table 4. Optimal control strategies of PVSDs for different orientations in Beijing.

City	Month	South	30 Degrees West/East of South	60 Degrees West/East of South	East/West
Beijing	January	60°~70°	60°~70°	60°~80°	60°~80°
	February	60°~70°	50°~70°	60°~70°	60°~80°
	March	50°~65°	50°~60°	50°~70°	50°~70°
	April	40°~50°	40°~50°	40°~60°	50°~70°
	May	40°~50°	30°~40°	40°~50°	40°~60°
	June	40°~50°	30°~40°	40°~50°	40°~60°
	July	40°~50°	30°~40°	40°~50°	40°~60°
	August	40°~50°	30°~40°	40°~50°	40°~60°
	September	50°~60°	40°~50°	50°~60°	50°~70°
	October	60°~70°	50°~60°	50°~70°	60°~70°
	November	60°~80°	50°~65°	60°~80°	60°~80°
	December	60°~80°	60°~70°	60°~80°	60°~80°

4. Conclusions

In this paper, a clear-sky solar radiation model is established to construct a numerical simulation study of PVSDs. Due to the isotropic nature of diffuse solar radiation, maximizing the collection of direct solar radiation is of great significance. And in this article, the influence of latitude and building orientation is analyzed quantitatively, which can cope with the integrated utilization of PVSDs for any building orientation facade in different latitude regions, thereby providing valuable guidance. The main conclusions of this study are summarized as follows:

(1) Across all latitudes, the higher the latitude, the higher the optimal tilt angle in the whole year, and the more significant increase in winter. For south-facing facades, the optimal tilt angle increases by 10° for every 10° increase in latitude in winter. And for every 25° increase in latitude, the optimal tilt angle increases by only about 10° in summer.

(2) The orientation of the building is a critical factor influencing the effectiveness of solar radiation utilization in PVSDs, and the efficiency markedly diminishes as the angle deviates from the due south direction. At the same time, under clear sky conditions, the optimal inclination angle difference between east and west directions is minimal, demonstrating precise symmetry.

(3) Compared to horizontal installation, vertical installation and annual optimal fixed tilt angle control strategies, the monthly optimal control strategy consistently achieves the highest solar radiation utilization efficiency. Furthermore, the efficiency disparities among the four regulation strategies become more pronounced as the orientation approaches due south. Due to the south orientation, the difference between the monthly optimal tilt strategy and the solar radiation utilization efficiency under the horizontal installation strategy is 34%, and the efficiency difference under the due east direction is 15%. Through the design of "optimal tilt angle interval for each month", a monthly regulation strategy is proposed that ensures the efficiency of solar radiation utilization exceeds 98% of the optimal tilt angle. This can be achieved through manual or straightforward mechanical adjustments once a month or quarterly, significantly lowering the operational threshold for PVSDs.

(4) The development of parametric performance design has established a visual design workflow, allowing for the calculation and assessment of the optimal inclination angle of PVSDs under various conditions without the need for specialized software or expertise such as EnergyPlus. This advancement enhances both the efficiency and universality of the design process.

(5) The design process of the optimal regulation strategy, encompassing geographical location, building orientation, and other influencing factors, has been established. It

can perform calculations and efficiency assessments of the optimal tilt angle range of PVSDs under various conditions, thereby improving design efficiency and increasing the applicability of PVSDs.

(6) In further research, the impact of natural daylighting on indoor lighting energy consumption will be considered and the optimal tilt angle will be investigated with the aim of minimizing the overall energy consumption of the buildings.

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Appendix A

Table A1. Optimal control strategies of PVSDs for different orientations across latitudes.

City	Month	South	30 Degrees West/East of South	60 Degrees West/East of South	East/West
Haikou	January	40°~60°	40°~60°	40°~60°	50°~70°
	February	40°~60°	40°~50°	40°~60°	50°~70°
	March	30°~45°	30°~45°	30°~50°	40°~60°
	April	20°~40°	30°~45°	30°~50°	30°~50°
	May	20°~40°	30°~45°	30°~50°	30°~50°
	June	20°~40°	30°~45°	30°~50°	30°~50°
	July	20°~40°	30°~45°	30°~50°	30°~50°
	August	20°~40°	30°~45°	30°~50°	30°~50°
	September	20°~40°	30°~45°	30°~50°	40°~60°
	October	30°~45°	40°~50°	40°~60°	40°~60°
	November	40°~60°	40°~50°	40°~60°	50°~65°
	December	40°~60°	40°~60°	50°~60°	50°~70°
Kunming	January	50°~65°	50°~60°	50°~70°	50°~70°
	February	40°~60°	40°~60°	40°~55°	50°~70°
	March	30°~50°	40°~50°	40°~55°	40°~60°
	April	30°~40°	30°~40°	30°~50°	40°~60°
	May	30°~40°	30°~40°	30°~50°	30°~50°
	June	30°~40°	30°~40°	30°~50°	30°~50°
	July	30°~40°	30°~40°	30°~50°	30°~50°
	August	30°~40°	30°~40°	30°~50°	30°~50°
	September	30°~40°	30°~50°	50°~60°	40°~60°
	October	40°~50°	40°~60°	40°~60°	50°~70°
	November	50°~60°	40°~50°	40°~65°	50°~70°
	December	50°~65°	50°~60°	50°~70°	60°~70°

Table A1. *Cont.*

City	Month	South	30 Degrees West/East of South	60 Degrees West/East of South	East/West
Nanjing	January	55°~70°	50°~70°	60°~70°	60°~70°
	February	50°~60°	50°~60°	50°~65°	60°~70°
	March	40°~50°	40°~50°	40°~60°	50°~65°
	April	30°~40°	30°~45°	40°~60°	40°~50°
	May	30°~40°	30°~45°	30°~50°	40°~50°
	June	30°~40°	30°~45°	30°~50°	40°~50°
	July	30°~40°	30°~45°	30°~50°	40°~50°
	August	30°~40°	30°~45°	40°~50°	40°~50°
	September	40°~50°	40°~50°	40°~50°	50°~60°
	October	50°~60°	40°~60°	50°~70°	60°~70°
	November	50°~70°	50°~70°	50°~70°	60°~80°
	December	60°~70°	50°~70°	50°~70°	60°~80°
Beijing	January	60°~70°	60°~70°	60°~80°	60°~80°
	February	60°~70°	50°~70°	60°~70°	60°~80°
	March	50°~65°	50°~60°	50°~70°	50°~70°
	April	40°~50°	40°~50°	40°~60°	50°~70°
	May	40°~50°	30°~40°	40°~50°	40°~60°
	June	40°~50°	30°~40°	40°~50°	40°~60°
	July	40°~50°	30°~40°	40°~50°	40°~60°
	August	40°~50°	30°~40°	40°~50°	40°~60°
	September	50°~60°	40°~50°	50°~60°	50°~70°
	October	60°~70°	50°~60°	50°~70°	60°~70°
	November	60°~80°	50°~65°	60°~80°	60°~80°
	December	60°~80°	60°~70°	60°~80°	60°~80°
Harbin	January	60°~80°	65°~80°	60°~80°	70°~80°
	February	55°~70°	60°~70°	60°~80°	60°~80°
	March	45°~60°	50°~60°	50°~70°	60°~70°
	April	40°~55°	40°~50°	40°~60°	50°~60°
	May	30°~50°	30°~50°	40°~60°	40°~60°
	June	30°~50°	30°~50°	40°~60°	40°~60°
	July	30°~50°	30°~50°	40°~60°	40°~60°
	August	40°~55°	40°~50°	40°~60°	40°~60°
	September	45°~60°	50°~60°	50°~60°	50°~70°
	October	55°~70°	60°~70°	50°~60°	60°~80°
	November	60°~80°	60°~80°	60°~80°	70°~80°
	December	60°~80°	65°~80°	60°~80°	70°~80°

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